

Data Flow Diagram

Date	26 June 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

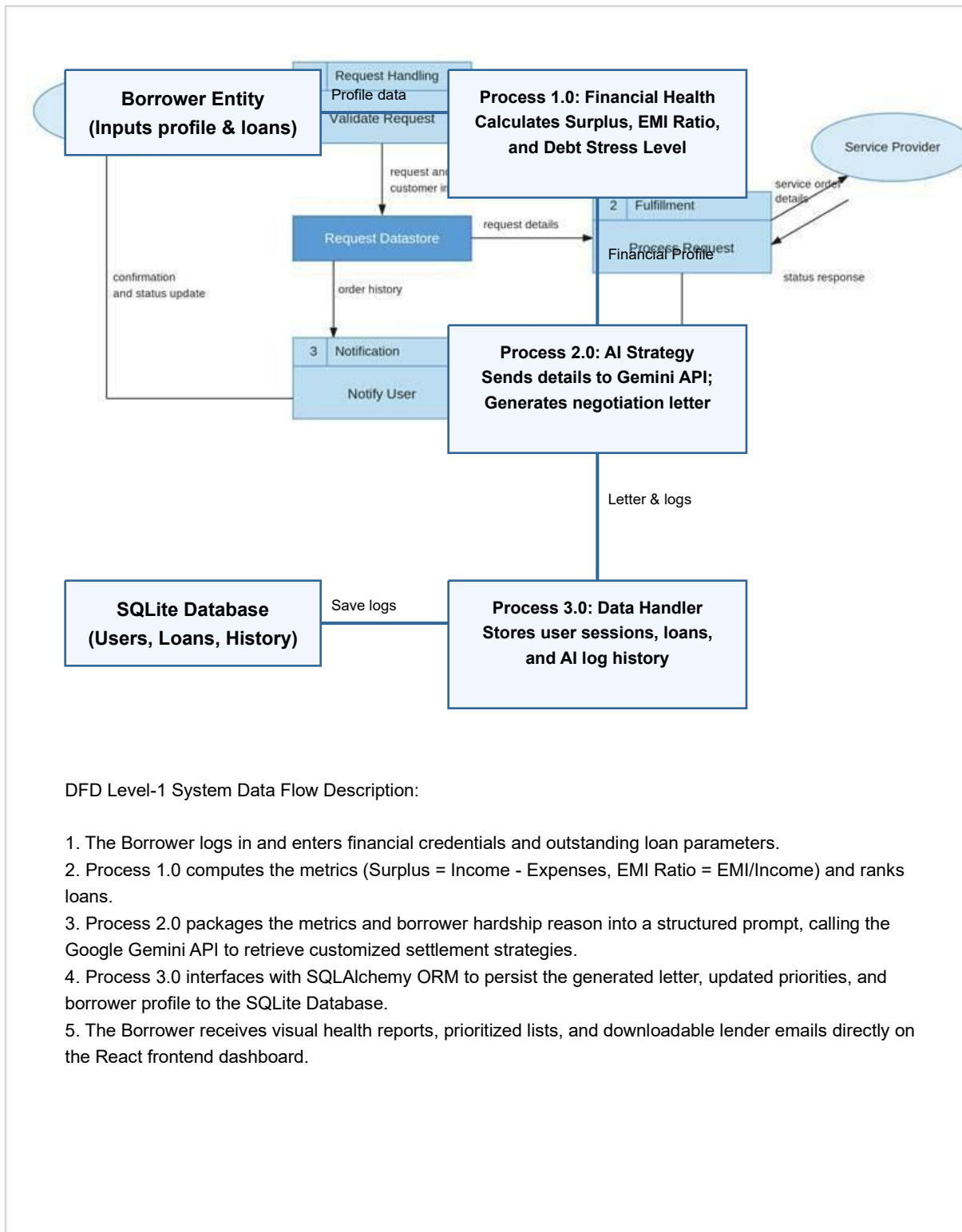
Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data(e.g., Validate Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example : The below diagram illustrates how the symbols above came together to form complete data flow diagram.



DFD Level-1 System Data Flow Description:

1. The Borrower logs in and enters financial credentials and outstanding loan parameters.
2. Process 1.0 computes the metrics ($\text{Surplus} = \text{Income} - \text{Expenses}$, $\text{EMI Ratio} = \text{EMI}/\text{Income}$) and ranks loans.
3. Process 2.0 packages the metrics and borrower hardship reason into a structured prompt, calling the Google Gemini API to retrieve customized settlement strategies.
4. Process 3.0 interfaces with SQLAlchemy ORM to persist the generated letter, updated priorities, and borrower profile to the SQLite Database.
5. The Borrower receives visual health reports, prioritized lists, and downloadable lender emails directly on the React frontend dashboard.